Automated Reasoning and In-Worker Verification for High-Rank Elliptic Curves: The 4-Stage Werr Algebraic Sieve, Conductor Certification ($\log N$), and Vélu Isogeny Graph Transport on the NSF ICARM Leaderboard

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Abstract

The computational verification of high-rank elliptic curves over $\mathbb{Q}$ has historically been bottlenecked by the difficulty of certifying global arithmetic invariants, most notably the arithmetic conductor N. In modern automated reasoning platforms such as the NSF Institute for Computer-Aided Reasoning in Mathematics (ICARM) Elliptic Curve Rank Leaderboard (maintained at Carnegie Mellon University under U.S. National Science Foundation Grant DMS-2425401), candidate curves are subjected to strict client-side and in-worker WebAssembly PARI/GP 2.17.2 kernel verification. While geometric non-singularity, rational point membership, torsion subgroup structures, and Néron–Tate height regulator determinants are verified efficiently, computing the global conductor via Tate's algorithm (`ellglobalred`) requires the complete prime factorization of the minimal discriminant Δ. For state-of-the-art curves specializing high-rank parametric families, $|\Delta|$ routinely spans 50 to 110 decimal digits, leaving many landmark curves across the literature with uncertified conductors (marked as `?` or unranked).

In this paper, we present an automated certificate-generation and isogeny graph exploration framework that resolves this bottleneck. Our methodology unites two engines tailored specifically to the arithmetic of elliptic curves: (1) the *4-Stage Werr Multi-Layer Algebraic Sieve*, which couples Stage 1 algebraic invariant-root and division-polynomial gcd kernels ($\gcd(c_4, c_6, \Delta, a_4, a_6)$), Stage 2 elliptic syzygy and monic cubic divisor peeling ($c_4^3 - 1728\Delta = c_6^2$, $\Delta = d_0 \cdot s^3$, and 2-torsion pairwise root-difference decomposition), Stage 3 parallel Pollard ρ, Pollard $p-1$, and multi-curve Montgomery/Weierstrass ECM ($B_1 = 10^5$, $B_2 = 5 \times 10^6$), and Stage 4 word-packed $\mathbb{F}_2$ Multipolynomial Quadratic Sieve (MPQS) with deterministic APR-CL / Miller–Rabin primality closure; and (2) systematic *Vélu Isogeny Graph Transport*, traversing rational n-isogeny neighborhoods ($n \in \{2, 3, 4, 5, 6, 8, 10, 12\}$) and exact point-interpolation Weierstrass recovery to discover minimal models with strictly reduced discriminants, Faltings heights, and naive heights while preserving Mordell–Weil rank and conductor.

Deploying this pipeline on the NSF ICARM leaderboard, we established **230 newly certified curves** (`#4308`–`#4537`) across 8 rational torsion subgroups with **100% certified conductor factorizations** (230/230, eliminating all missing conductor certificates), securing **30 #1 Gold records**, **80 Top-3 podium finishes**, and **418 Top-10 placements** across 43 active leaderboards. Landmark results include the "Grand Slam" curve `#4315` ($T = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, Rank ≥ 9), which captures `#1` in all four metrics simultaneously, the factorization of a 110-digit discriminant into 21 distinct bad primes for the Rank ≥ 12 curve `#4383`, and full conductor certification ($\log N = 141.86$) for the Elkies Rank ≥ 19 curve `#4323`. All datasets, reproduction scripts, and verification receipts are permanently archived at CERN Zenodo (DOI: 10.5281/zenodo.23436282).

Keywords: Automated Reasoning, Proof Certificates, Elliptic Curves, Conductor Certification, Tate's Algorithm, Vélu Isogenies, Algebraic Sieve, Symbolic Computation, Computer-Aided Mathematics.

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1 Introduction and Computer-Aided Reasoning in Arithmetic Geometry

The interface between automated deduction, formal proof verification, and computational number theory has undergone a profound transformation over the past decade [1,2,8]. Historically, computational records in arithmetic geometry—such as tables of elliptic curves over $\mathbb{Q}$ with high Mordell–Weil rank or prescribed rational torsion subgroups—were compiled as static text repositories maintained by individual researchers. While foundational to number theory, static tables rely on implicit trust in unbundled off-line scripts and frequently omit computationally expensive global invariants.

A major advance toward interactive, machine-certified mathematical benchmarking is the **NSF Institute for Computer-Aided Reasoning in Mathematics (ICARM)**, established at Carnegie Mellon University and partner institutions under U.S. National Science Foundation Grant DMS-2425401. As part of its automated reasoning infrastructure, ICARM operates the official **Elliptic Curve Rank Leaderboard** (`https://elliptic-rank.icarm.cloud`), architected and maintained by David Renshaw and Seewoo Lee [9]. Rather than accepting unverified user assertions, the ICARM platform enforces a zero-trust, certificate-driven verification protocol executed inside an isolated WebAssembly (WASM) worker running the computer algebra kernel **PARI/GP 2.17.2** [12].

1.1 The In-Worker WebAssembly Verification Protocol

When a contributor submits a candidate elliptic curve $E/\mathbb{Q}$ defined by integer Weierstrass coefficients $\mathbf{a} = [a_1, a_2, a_3, a_4, a_6]$, a list of rational point coordinates $\mathcal{P} = \{P_1, \dots, P_r\}$, and an optional list of prime factors $\mathcal{S} = \{p_1, \dots, p_k\}$, the ICARM verification kernel executes a six-stage formal decision procedure:

1. **Geometric Non-Singularity Check:** The kernel evaluates the Tate invariants b_2, b_4, b_6, b_8, the covariants c_4, c_6, and the discriminant $\Delta(\mathbf{a})$. It verifies that $\Delta(\mathbf{a}) \neq 0$, guaranteeing that E defines a smooth projective curve of genus 1 over $\mathbb{Q}$.
2. **Canonical Isomorphism Key Generation:** Using LMFDB-compatible invariant reduction, the kernel removes admissible scaling powers $u = 2^\alpha 3^\beta$ from (c_4, c_6) to compute the canonical isomorphism class key `"c4:c6"`, preventing duplicate submissions of isomorphic models.
3. **Exact Rational Point Membership:** For every asserted generator $P_j = (x_j, y_j) \in \mathbb{Q}^2$, the kernel verifies exact arithmetic equality in $\mathbb{Q}$:
$$y_j^2 + a_1 x_j y_j + a_3 y_j = x_j^3 + a_2 x_j^2 + a_4 x_j + a_6. \tag{1}$$
4. **Torsion Subgroup Certification:** Applying Lutz–Nagell bounds and division polynomial root analysis via `elltors(E)`, the kernel certifies the exact rational torsion subgroup $E(\mathbb{Q})_{\text{tors}} \cong T$ among the 15 admissible groups classified by Mazur's Torsion Theorem [11].
5. **Mordell–Weil Rank Certification via Regulator Positivity:** The kernel computes the canonical Néron–Tate height pairing matrix
$$R(\mathcal{P}) = (\langle P_i, P_j \rangle)_{1 \le i,j \le r}, \quad \langle P_i, P_j \rangle = \hat{h}(P_i + P_j) - \hat{h}(P_i) - \hat{h}(P_j), \tag{2}$$
using `ellbil(E, P_i, P_j)` at high precision, and extracts a maximal $\mathbb{Z}$-linearly independent subset of $\mathcal{P}$ whose Gram determinant $\det R > 0$ is strictly bounded away from zero. This provides a machine-certified lower bound $\text{rank}(E(\mathbb{Q})) \ge r$.
6. **Global Minimal Reduction and Conductor Certification ($\log N$):** Finally, the kernel invokes Tate's global reduction algorithm [15] via `ellglobalred(E)` to transform E to its global minimal integral Weierstrass equation, compute the minimal discriminant $|\Delta_{\min}|$, the Faltings height $h_F(E)$, the naive height $h_{\text{naive}}(E)$, and the global arithmetic conductor N.

1.2 The Conductor Bottleneck on High-Rank Leaderboards

Across the ICARM leaderboard, curves within each (T, r) category—where $T = E(\mathbb{Q})_{\text{tors}}$ is the torsion subgroup and r is the certified Mordell–Weil rank lower bound—are ranked across four primary minimality metrics:

$$\text{(i) Conductor } N, \quad \text{(ii) Discriminant } |\Delta|, \quad \text{(iii) Faltings Height } h_F(E), \quad \text{(iv) Naive Height } h_{\text{naive}}(E). \tag{3}$$

While $|\Delta|$, $h_F(E)$, and $h_{\text{naive}}(E)$ can be evaluated directly from the Weierstrass coefficients once local scaling factors at $p \in \{2, 3\}$ are removed, computing the conductor $N = \prod_{p|\Delta} p^{f_p}$ requires knowing the *complete prime factorization* of the discriminant Δ.

In high-rank elliptic curve constructions pioneered by Noam Elkies [7], Andrej Dujella [6], Michael Rubinstein [13], and Zev Klagsbrun, curves of rank $r \in [7, 29]$ are obtained by specializing multi-parameter K3 and rational elliptic surfaces over $\mathbb{Q}(t_1, \ldots, t_m)$ at rational parameters of large height. Because the discriminant Δ is a polynomial of high degree in the specialization parameters, the resulting integer $|\Delta|$ routinely spans **50 to 110 decimal digits** (165 to 365 bits). When submitted without a pre-computed prime factorization certificate $\mathcal{S}$, the in-worker PARI/GP kernel cannot factor $|\Delta|$ within the interactive execution budget, leaving the conductor field marked as `?` (unfactored/unknown) and disqualifying the curve from the Conductor N leaderboard.

Furthermore, many high-rank curves in the historical literature are recorded in a single Weierstrass coordinate system tied to their fibration parameterization, without exploring the finite graph of $\mathbb{Q}$-isogenous models. Because rational isogenies preserve both the Mordell–Weil rank r and the conductor N while altering $|\Delta|$, $h_F(E)$, and $h_{\text{naive}}(E)$, uninspected vertices of the rational isogeny graph often conceal strictly smaller minimal models.

1.3 Summary of Contributions

In this work, we bridge automated proof-certificate generation and computational arithmetic geometry through three primary contributions:

1. **The 4-Stage Werr Multi-Layer Algebraic Sieve for Elliptic Discriminants:** We design a specialized four-stage factorization and certificate synthesis pipeline that exploits Weierstrass covariant gcds, elliptic syzygy cubic peeling ($\Delta = d_0 \cdot s^3$), 2-torsion monic cubic root-difference splitting, parallel Pollard $\rho/p-1$ and multi-curve ECM, and word-packed $\mathbb{F}_2$ MPQS closure to completely factor composite discriminants up to 110 decimal digits.
2. **Systematic Vélu Isogeny Graph Transport and Point-Interpolation Recovery:** We implement an exact rational isogeny graph explorer covering subgroups of orders $n \in \{2, 3, 4, 5, 6, 7, 8, 9, 10, 12\}$ together with an exact linear point-interpolation solver that reconstructs missing Weierstrass models from historical point tables and transports Mordell–Weil generators $P_j \mapsto \phi(P_j)$ across 2- and 3-isogeny graphs.
3. **Comprehensive Public Certification on the NSF ICARM Leaderboard:** We submitted and certified **230 distinct Top-10 curves** (`#4308` through `#4537`) under contributor profile `user/114` (https://elliptic-rank.icarm.cloud/user/114), achieving **100.0% conductor certification** (230/230 with zero unknown conductors), **30 #1 Gold records**, **80 Top-3 podiums**, and **418 Top-10 metric placements** across 43 active leaderboards [5].

2 Conductor Determination as Formal Proof Certificate Checking

Let $E/\mathbb{Q}$ be an elliptic curve defined by a generalized Weierstrass equation with integer coefficients:

$$E: \quad y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \quad a_1, a_2, a_3, a_4, a_6 \in \mathbb{Z}. \tag{4}$$

The standard Tate quantities b_2, b_4, b_6, b_8 and covariants c_4, c_6, Δ are given by:

$$b_2 = a_1^2 + 4a_2, \qquad b_4 = 2a_4 + a_1a_3, \qquad b_6 = a_3^2 + 4a_6, \tag{5}$$

$$b_8 = a_1^2a_6 + 4a_2a_6 - a_1a_3a_4 + a_2a_3^2 - a_4^2, \tag{6}$$

$$c_4 = b_2^2 - 24b_4, \qquad c_6 = -b_2^3 + 36b_2b_4 - 216b_6, \tag{7}$$

$$\Delta = -b_2^2b_8 - 8b_4^3 - 27b_6^2 + 9b_2b_4b_6 = \frac{c_4^3 - c_6^2}{1728}. \tag{8}$$

An admissible change of Weierstrass coordinates over $\mathbb{Q}$ takes the form $(x, y) = (u^2x' + r, u^3y' + u^2sx' + t)$ with $u \in \mathbb{Q}^\times$ and $r, s, t \in \mathbb{Q}$, under which the invariants transform as:

$$c_4' = u^{-4}c_4, \qquad c_6' = u^{-6}c_6, \qquad \Delta' = u^{-12}\Delta. \tag{9}$$

An integral model (4) is *globally minimal* over $\mathbb{Z}$ if $v_p(\Delta)$ is minimized simultaneously at every prime p. Because $\mathbb{Z}$ has class number 1, every elliptic curve $E/\mathbb{Q}$ admits a unique normalized global minimal model satisfying $a_1, a_3 \in \{0, 1\}$ and $a_2 \in \{-1, 0, 1\}$.

Definition 2.1 (Arithmetic Conductor and Ogg–Saito Formula)**.** Let $E/\mathbb{Q}$ be a globally minimal Weierstrass model with minimal discriminant $\Delta_{\min}$. The *arithmetic conductor* of E is the positive integer

$$N(E) = \prod_{p \mid \Delta_{\min}} p^{f_p}, \tag{10}$$

where the local exponent $f_p \in \mathbb{Z}_{\geq 0}$ is given by the Ogg–Saito formula:

$$f_p = v_p(\Delta_{\min}) - m_p + 1, \tag{11}$$

with m_p denoting the number of irreducible components (counted without multiplicity) of the special fiber $\mathcal{E}_p$ of the Néron minimal model of E over $\mathbb{Z}_p$. Specifically:

- $f_p = 0$ if and only if $p \nmid \Delta_{\min}$ (E has good reduction at p, Kodaira type I_0);
- $f_p = 1$ if and only if $p \mid \Delta_{\min}$ and $p \nmid c_4$ (E has multiplicative reduction at p, Kodaira type I_n with $n = v_p(\Delta_{\min}) \geq 1$);
- $f_p = 2 + \delta_p \geq 2$ if E has additive reduction at p ($p \mid \gcd(c_4, \Delta_{\min})$), where the wild Swan ramification invariant $\delta_p = 0$ for all primes $p \geq 5$, while $0 \leq \delta_3 \leq 3$ ($f_3 \leq 5$) and $0 \leq \delta_2 \leq 6$ ($f_2 \leq 8$).

Proposition 2.2 (Certificate Verification Complexity of Tate's Algorithm)**.** *Let $E/\mathbb{Q}$ be specified by an integral Weierstrass tuple $\mathbf{a} = [a_1, a_2, a_3, a_4, a_6] \in \mathbb{Z}^5$ with discriminant $\Delta \neq 0$. Suppose an external prover supplies a prime factorization certificate*

$$\mathcal{S} = \{p_1, p_2, \ldots, p_k\} \quad \textit{such that} \quad |\Delta| = \prod_{i=1}^{k} p_i^{e_i}, \quad e_i = v_{p_i}(\Delta) \geq 1. \tag{12}$$

Then a verification kernel can certify the primality of each p_i, execute Kraus–Laska–Connell local minimization at every p_i with $e_i \geq 12$, run Tate's algorithm [15] at all bad primes, and output the exact global minimal model $E_{\min}$ and certified conductor $N(E)$ in deterministic time polynomial in $\log|\Delta|$.

Proof. Verification of the identity $|\Delta| = \prod_{i=1}^{k} p_i^{e_i}$ requires at most $\sum_{i=1}^{k} e_i \leq \log_2|\Delta|$ multi-precision divisions of bit-length $B = \lceil \log_2|\Delta| \rceil$, costing $O(B^2)$ bit operations. Primality verification of each $p_i \leq |\Delta|$ via deterministic APR-CL [3] or ECPP certificates runs in time $O((\log p_i)^{4+\varepsilon})$. For each prime $p_i \in \mathcal{S}$, Tate's algorithm [4, 15] executes at most $\lfloor e_i/12 \rfloor + 1$ local loop iterations. Each iteration computes p_i-adic valuations of the invariants and finds roots over $\mathbb{F}_{p_i}$ of auxiliary polynomials of degree at most 3:

$$P_1(T) = T^2 + a_1 T - a_2 \pmod{p_i}, \tag{13}$$

$$P_2(Y) = Y^2 + a_3 Y - a_6 \pmod{p_i}, \tag{14}$$

$$P_3(X) = X^3 + a_{2,1}X^2 + a_{4,2}X + a_{6,3} \pmod{p_i}, \quad a_{j,m} = p_i^{-m} a_j. \tag{15}$$

Root-finding for degree ≤ 3 polynomials over $\mathbb{F}_{p_i}$ requires $O(\log^2 p_i)$ field operations via Cantor–Zassenhaus. Since $k \leq \log_2|\Delta|$ and $\sum_{i=1}^{k} \log p_i \leq \log|\Delta|$, the entire global reduction and conductor certification procedure terminates in $O(\mathrm{poly}(\log|\Delta|))$ time. □

Remark 2.3 (Automated Reasoning Architecture: Separating Search from Verification). Proposition 2.2 exemplifies the foundational design principle of modern computer-aided reasoning (`cs.LO`): *separating untrusted, computationally intensive certificate search from lightweight, deterministic kernel verification.* In the ICARM architecture, the submission payload includes the field `"primes": ["p_1", ..., "p_k"]`. Crucially, if a submitted model $\mathbf{a}$ is non-minimal with scaling factor $u \in \mathbb{Z}_{>1}$, its raw discriminant satisfies $\Delta(\mathbf{a}) = u^{12}\Delta_{\min}$. For PARI/GP's in-worker `ellglobalred` routine to succeed without factoring timeouts, the certificate $\mathcal{S}$ must cover all prime divisors of the *submitted* discriminant $\Delta(\mathbf{a})$ (including primes dividing u). Once $\mathcal{S}$ is registered via `addprimes(S)`, the WebAssembly worker verifies trial-division completeness against $\Delta(\mathbf{a})$, strips u^{12}, and certifies $\log N$ in under 2 seconds.

3 The 4-Stage Werr Multi-Layer Algebraic Sieve

For high-rank elliptic curves with 70 to 110 digit discriminants, general-purpose integer factorization of Δ as an unstructured black-box integer is frequently intractable within interactive time limits. To overcome this, we developed the **4-Stage Werr Multi-Layer Algebraic Sieve**, documented in our Zenodo research monograph [5] and summarized in Table 1.

3.1 Stage 1: Algebraic Root and Covariant GCD Kernel

Before invoking probabilistic or sub-exponential factoring routines, Stage 1 exploits the polynomial dependencies among the Weierstrass coefficients and Tate covariants. Specifically, Stage 1 strips all small primes $p \leq 10^5$ via a fast byte-sieve wheel (`SMALL_PRIMES`) and computes the mutual greatest common divisors:

$$g_1 = \gcd(|c_4|, |c_6|), \quad g_2 = \gcd(|a_4|, |a_6|), \quad g_3 = \gcd(|b_2|, |b_4|, |b_6|, |b_8|), \quad g_4 = \gcd(|\Delta|, |c_4|, |c_6|). \tag{16}$$

Table 1: Architecture of the 4-Stage Werr Multi-Layer Algebraic Sieve for Elliptic Curve Discriminants

Pipeline Stage	**Algebraic Identity / Method**	**Target Scale**	**Mechanism & Role in Conductor Certification**
Stage 1 Algebraic Root & GCD Kernel	$\gcd(c_4, c_6, \Delta, a_4, a_6, b_2, b_4, b_6)$ Division polynomials $\psi_2(x), \psi_3(x)$	Smooth towers & algebraic roots	Extracts common polynomial divisor towers across Weierstrass invariants and locates exact integer roots of monic 2-torsion cubics and 3-torsion quartics via binary/Newton root bracketing.
Stage 2 Cubic & Syzygy Divisor Peeling	$c_4^3 - 1728\Delta = c_6^2 \implies \Delta = d_0 \cdot s^3$ $\Delta = 2^{-12}\prod_{i<j}(X_i - X_j)^2$	40–110 digit structured Δ	Isolates cubic divisor towers $s^3 \mid \gcd(c_4^3, \Delta)$, strips non-minimal scaling factors u^{12}, and decomposes 2-torsion discriminants into squares of pairwise root differences $\lvert X_i - X_j\rvert$.
Stage 3 Parallel Pollard ρ, $p-1$ & ECM	Batched Brent ρ (2.5×10^5 steps) Montgomery/Weierstrass ECM ($B_1 = 10^5, B_2 = 5 \times 10^6$)	12–32 digit prime factors	Multi-core worker threads sweep residual composite cofactors in parallel using Montgomery/Weierstrass elliptic curve group law additions modulo C to extract medium-to-large prime factors.
Stage 4 Word-Packed $\mathbb{F}_2$ MPQS & APR-CL	Multipolynomial Quadratic Sieve 64-bit word-packed $\mathbb{F}_2$ Gaussian PARI/GP 2.17.2 APR-CL closure	60–105 digit composites (18–38d primes)	Splits remaining balanced composite cofactors via word-packed $\mathbb{F}_2$ nullspace elimination and certifies all prime factors via deterministic APR-CL and Baillie–PSW tests for `ellglobalred`.

Whenever $g_k > 1$, Euclidean gcd splitting immediately factors Δ into smaller coprime components Δ/g_k^m and g_k. Furthermore, when E possesses non-trivial rational torsion, the division polynomials $\psi_n(x)$ impose exact polynomial factorizations on Δ:

Proposition 3.1 (Algebraic Discriminant Decomposition via Rational 2-Torsion)**.** *Let $E/\mathbb{Q}$ be an integral Weierstrass model possessing at least one rational 2-torsion point $P_0 = (x_0, y_0) \in E(\mathbb{Q})[2] \setminus \{O\}$. Then $X_0 = 4x_0 \in \mathbb{Z}$ is an integer root of the monic cubic polynomial*

$$f(X) = X^3 + B_2X^2 + B_4X + B_6 = 0, \quad B_2 = b_2, \quad B_4 = 8b_4, \quad B_6 = 16b_6 \in \mathbb{Z}. \tag{17}$$

Factoring $f(X) = (X - X_0)(X^2 + \alpha X + \beta)$ with $\alpha = B_2 + X_0 \in \mathbb{Z}$ and $\beta = B_4 + \alpha X_0 \in \mathbb{Z}$ decomposes the elliptic discriminant $\Delta(E)$ algebraically into the product:

$$\Delta(E) = 2^{-12} \cdot (\alpha^2 - 4\beta) \cdot (3X_0^2 + 2B_2X_0 + B_4)^2. \tag{18}$$

In particular, if $E(\mathbb{Q})[2] \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ with three integer roots X_0, X_1, X_2 of $f(X)$, then $\Delta(E)$ splits completely into the square of the pairwise root differences:

$$\Delta(E) = 2^{-12} \cdot (X_0 - X_1)^2(X_0 - X_2)^2(X_1 - X_2)^2. \tag{19}$$

Proof. A point $P_0 = (x_0, y_0)$ satisfies $2P_0 = O$ if and only if $2y_0 + a_1x_0 + a_3 = 0$. Completing the square in the Weierstrass equation (4) yields $(2y + a_1x + a_3)^2 = 4x^3 + b_2x^2 + 2b_4x + b_6$. Multiplying by 16 and substituting $X = 4x$ produces the monic integer polynomial $f(X) = X^3 + B_2X^2 + B_4X + B_6$. By Gauss's Lemma, any rational root of $f(X)$ is an integer $X_0 \in \mathbb{Z}$. The polynomial discriminant of $f(X)$ is $\mathrm{Disc}(f) = 2^{12}\Delta(E)$. For a factored polynomial $f(X) = (X - X_0)q(X)$, the resultant identity $\mathrm{Disc}(f) = \mathrm{Disc}(q) \cdot q(X_0)^2 = (\alpha^2 - 4\beta)(f'(X_0))^2$ yields (18) and (19). □

Crucially, finding the integer roots $X_0 \in \mathbb{Z}$ of $f(X)$ requires *zero integer factorization*: since $f(X)$ is a cubic with at most two real local extrema, Stage 1 computes the overflow-safe bit-length bound $M = 2^{\max(\mathrm{bits}(B_2), \mathrm{bits}(B_4)/2, \mathrm{bits}(B_6)/3)+4}$ and locates all integer roots in $O(\log |B_6|)$ arithmetic operations via exact integer bisection (`integer_roots_monic_cubic`). For curves with full 2-torsion $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (such as the 110-digit discriminant curve `#4383`), identity (19) immediately reduces the problem of factoring a 110-digit integer $|\Delta|$ to factoring three integers $|X_i - X_j|$ of at most 18–20 digits each!

3.2 Stage 2: Elliptic Syzygy and Cubic Divisor Peeling

For curves with trivial 2-torsion ($E(\mathbb{Q})[2] = \{O\}$), Stage 2 exploits the fundamental elliptic syzygy:

$$c_4^3 - 1728\Delta = c_6^2 \implies c_4^3 \equiv c_6^2 \pmod{1728\Delta}. \tag{20}$$

Any prime $p \geq 5$ dividing both c_4 and Δ (primes of additive reduction) satisfies $p^3 \mid c_4^3$ and $p^2 \mid c_6^2$, allowing Stage 2 to isolate cubic and square divisor factors $\Delta = d_0 \cdot s^3$ via iterated $\gcd(\Delta, c_4^3)$ and exact integer root extraction `isqrt`($curr$). Similarly, when an unreduced or Vélu-transported model has $\Delta_{\mathrm{sub}} = u^{12}\Delta_{\mathrm{min}}$, Stage 2 peels the 12th-power scaling factor u^{12} by computing $\gcd(c_4^3, c_6^2, \Delta)$.

3.3 Stage 3: Parallel Pollard ρ, Pollard $p-1$, and Multi-Curve ECM

Each remaining composite cofactor $C > 1$ (after checking deterministic primality via `is_probable_prime` and exact square peeling `isqrt`) is pushed onto a factorization work-stack and processed by Stage 3:

1. **Batched Pollard–Brent ρ:** Workers execute Brent's cycle-finding improvement to Pollard's ρ algorithm with polynomial transition $x \mapsto (x^2 + c) \bmod C$, accumulating modular products in batches of 128 steps before computing $\gcd(\prod |x_i - y_i|, C)$ up to 2.5×10^5 iterations. This rapidly strips all prime factors up to 14–16 decimal digits.
2. **Multi-Curve Weierstrass/Montgomery ECM (`ecm_stage1`):** For cofactors containing 15 to 32 digit primes, Stage 3 launches parallel Lenstra Elliptic Curve Method (ECM) workers [10]. Each worker generates random curves $E_a : y^2 = x^3 + ax + b \pmod C$ passing through a random initial point $P_0 = (x_0, y_0) \bmod C$ by setting $b \equiv y_0^2 - x_0^3 - ax_0 \pmod C$, checks $\gcd(4a^3 + 27b^2, C)$, and multiplies P_0 by prime powers $p^k \le B_1$ ($B_1 = 10^5$, Stage 2 bound $B_2 = 5 \times 10^6$). Whenever a projective denominator $2y_Q$ or $(x_Q - x_R)$ shares a non-trivial factor with C, the modular inverse exception $\gcd(\text{den}, C) \in (1, C)$ immediately yields a prime factor of C.

3.4 Stage 4: Word-Packed $\mathbb{F}_2$ MPQS and PARI/GP 2.17.2 Closure

When a composite cofactor C of 60 to 130 bits resists ECM (typically a balanced semiprime product of two 18 to 22 digit primes, as in the 74-digit discriminant of the Rank ≥ 19 curve `#4323`), Stage 4 invokes our Word-Packed $\mathbb{F}_2$ Multipolynomial Quadratic Sieve (`solve_word_packed_f2_qs`) and PARI/GP 2.17.2 MPQS closure:

- Polynomials $Q(x) = (Ax + B)^2 - C$ are sieved over the factor base $\mathcal{B} = \{p \le B_{\max} : (\frac{C}{p}) = 1\}$ using Tonelli–Shanks modular square roots.
- Smooth exponent vectors modulo 2 are packed into 64-bit machine words (`uint64`), and $\mathbb{F}_2$-nullspace dependencies are extracted via bitwise MSB-pivot Gaussian elimination to construct $X^2 \equiv Y^2 \pmod C$, yielding $\gcd(X - Y, C) = p$.
- Every extracted factor is certified prime via deterministic Miller–Rabin / Baillie–PSW and APR-CL primality tests, producing the sorted bad-prime certificate $\mathcal{S} = \{p_1, \ldots, p_k\}$ ready for ICARM API submission.

Table 2 presents concrete factorization results from our 230 certified curves, illustrating how the 4-stage pipeline resolves both 110-digit multi-factor discriminants and 30-to-38 digit prime factors at ranks $r \in [12, 19]$.

Table 2: Representative High-Rank Discriminant Factorizations Certified by the 4-Stage Werr Sieve

Curve ID	Torsion T	Rank	$\lvert\Delta\rvert$ Dig.	N Dig.	$\log N$	Primes	Largest Prime Factor $p_{\max}$ Dividing Δ	$p_{\max}$ Dig.
#4323	Trivial []	≥ 19	74 d	62 d	141.86	9	5925062485692821329 (3 primes: 18d, 18d, 19d)	19 d
#4395	Trivial []	≥ 15	39 d	38 d	87.49	2	197846159326450166819037158651976203 39	**38** d
#4393	Trivial []	≥ 15	38 d	38 d	86.59	3	549109652144331750922854086151 19	**32** d
#4394	Trivial []	≥ 15	39 d	38 d	87.49	4	84059818790811557180008009 9	27 d
#4397	Trivial []	≥ 14	34 d	34 d	76.85	3	92383824520523308071859650 3101	30 d
#4466	Trivial []	≥ 13	31 d	31 d	69.19	1	1122482466080847283271138480843 (Prime $\lvert\Delta\rvert = N$)	31 d
#4465	Trivial []	≥ 13	30 d	30 d	68.26	2	22100397352440688467718468 6823	30 d
#4383	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 12	**110** d	48 d	110.22	**21**	363119 (21 primes via Stage 1–2 cubic peeling)	6 d
#4321	$\mathbb{Z}/5\mathbb{Z}$	≥ 7	65 d	22 d	50.13	8	115925634521	12 d
#4320	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$	≥ 6	52 d	18 d	39.78	11	947	3 d
#4319	$\mathbb{Z}/4\mathbb{Z}$	≥ 8	50 d	21 d	48.33	11	6367	4 d
#4315	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 9	46 d	22 d	50.40	10	1433 (Grand Slam: `#1` in all 4 metrics)	4 d

4 Vélu Rational Isogeny Graph Transport and Minimal Model Search

While the 4-Stage Werr Algebraic Sieve resolves the conductor certification bottleneck ($\log N$), our second engine exploits the geometry of rational isogeny graphs to discover strictly smaller minimal Weierstrass models and populate adjacent torsion leaderboards.

4.1 Invariance of Rank and Conductor Across $\mathbb{Q}$-Isogeny Orbits

Let $E/\mathbb{Q}$ and $E'/\mathbb{Q}$ be elliptic curves connected by a non-constant $\mathbb{Q}$-rational isogeny $\phi : E \to E'$ of degree $n = |\ker \phi|$. Let $\hat{\phi} : E' \to E$ denote the dual isogeny satisfying $\hat{\phi} \circ \phi = [n]_E$ and $\phi \circ \hat{\phi} = [n]_{E'}$.

Theorem 4.1 (Invariance of Rank, Conductor, and L-Function under $\mathbb{Q}$-Isogeny)**.** *Let $\phi : E \to E'$ be a $\mathbb{Q}$-rational isogeny between elliptic curves over $\mathbb{Q}$. Then:*

1. ***Mordell–Weil Rank Invariance:*** *ϕ induces a $\mathbb{Q}$-linear isomorphism $E(\mathbb{Q}) \otimes_{\mathbb{Z}} \mathbb{Q} \xrightarrow{\sim} E'(\mathbb{Q}) \otimes_{\mathbb{Z}} \mathbb{Q}$, so* $\operatorname{rank}(E'(\mathbb{Q})) = \operatorname{rank}(E(\mathbb{Q}))$.
2. ***Conductor and Bad Prime Invariance:*** *By the Néron–Ogg–Shafarevich criterion and Tate's isogeny theorem, the ℓ-adic Galois representations $V_\ell(E) \cong V_\ell(E')$ are isomorphic as $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-modules. Consequently, E and E' have the exact same set of bad primes $\mathcal{S}$, the same global arithmetic conductor $N(E') = N(E)$, and identical Hasse–Weil L-functions $L(E', s) = L(E, s)$.*

Crucially, however, the *minimal discriminant* $|\Delta(E')|$, the *Faltings height* $h_F(E')$, and the *naive height* $h_{\text{naive}}(E') = \log \max(|c_4|^3, c_6^2)$ are **not** invariant under isogeny! For a prime ℓ-isogeny $\phi : E \to E'$, a multiplicative fiber of Kodaira type I_m at a prime $p \mid N$ transitions either to $\mathrm{I}_{m\cdot\ell}$ or to $\mathrm{I}_{m/\ell}$, multiplying or dividing $|\Delta_{\min}|$ by ℓ^m across multiple primes while shifting the Faltings height via Faltings' isogeny degree formula.

4.2 Explicit Vélu Quotient Equations and Generator Point Transport

Let $G \subset E(\mathbb{Q})$ be a finite $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$-stable subgroup of order n. Following Vélu [16], we partition the non-zero elements $G \setminus \{O\} = G_2 \sqcup G_+ \sqcup G_-$, where $G_2 = \{Q \in G \setminus \{O\} : 2Q = O\}$ consists of the 2-torsion points in G, and each inverse pair $\{Q, -Q\}$ of order > 2 has one representative in G_+ and one in G_-. Setting $S = G_2 \sqcup G_+$, for each $Q = (x_Q, y_Q) \in S$ we compute in exact rational arithmetic (`fractions.Fraction`):

$$g_Q^x = 3x_Q^2 + 2a_2x_Q + a_4 - a_1y_Q, \qquad g_Q^y = -2y_Q - a_1x_Q - a_3, \tag{21}$$

$$t_Q = \begin{cases} g_Q^x & \text{if } Q \in G_2, \\ 2g_Q^x - a_1g_Q^y & \text{if } Q \in G_+, \end{cases} \qquad u_Q = (g_Q^y)^2, \qquad w_Q = u_Q + x_Q t_Q. \tag{22}$$

Summing $t = \sum_{Q\in S} t_Q$ and $w = \sum_{Q\in S} w_Q$, the quotient curve $E' = E/G$ is given by the Weierstrass model $\mathbf{A} = [A_1, A_2, A_3, A_4, A_6]$:

$$A_1 = a_1, \quad A_2 = a_2, \quad A_3 = a_3, \quad A_4 = a_4 - 5t, \quad A_6 = a_6 - b_2t - 7w. \tag{23}$$

Simultaneously, for every independent Mordell–Weil generator $P_j = (x_j, y_j) \in \mathcal{P}$ on $E(\mathbb{Q})$, our engine transports P_j across the isogeny via the exact rational map $\phi(P_j) = (X_j, Y_j) \in E'(\mathbb{Q})$:

$$X_j = x_j + \sum_{Q\in G\setminus\{O\}} \left(x(P_j + Q) - x(Q)\right), \qquad Y_j = y_j + \sum_{Q\in G\setminus\{O\}} \left(y(P_j + Q) - y(Q)\right). \tag{24}$$

When a rational subgroup G of order 3 is generated by a root $x_0 \in \mathbb{Q}$ of the 3-division polynomial $\psi_3(x) = 3x^4 + b_2x^3 + 3b_4x^2 + 3b_6x + b_8 = 0$ whose y-coordinate y_0 lies in a quadratic extension $\mathbb{Q}(\sqrt{d})$, note that $t_Q = 2g_Q^x - a_1g_Q^y$ and $u_Q = (g_Q^y)^2$ depend only on x_0 and $(2y_0 + a_1x_0 + a_3)^2 = 4x_0^3 + b_2x_0^2 + 2b_4x_0 + b_6 \in \mathbb{Q}$! Thus, our 3-isogeny engine (`velu_3isogeny_from_x0`) evaluates both E/G and the transported rational points $\phi(P_j)$ purely over $\mathbb{Q}$ without ever instantiating $\mathbb{Q}(\sqrt{d})$.

When the coordinates of $Q \in G \setminus \{O\}$ have rational denominators (e.g., $x_0 = X_0/4$), our engine clears denominators via the minimal scaling factor $u = 2^\alpha 3^\beta$ such that

$$\tilde{A}_1 = uA_1, \quad \tilde{A}_2 = u^2A_2, \quad \tilde{A}_3 = u^3A_3, \quad \tilde{A}_4 = u^4A_4, \quad \tilde{A}_6 = u^6A_6 \in \mathbb{Z}, \tag{25}$$

and scales every transported generator accordingly via $(\tilde{X}_j, \tilde{Y}_j) = (u^2X_j, u^3Y_j)$.

4.3 Point-Interpolation Recovery and Cross-Torsion Isogeny Transitions

In several archival web pages (such as Dujella's historical subpages `z2z2old*.html` and `z6old*.html`), high-rank records are listed only by their independent generator points $\{P_1, \ldots, P_r\}$ without explicit Weierstrass coefficients $[a_1, a_2, a_3, a_4, a_6]$. Our pipeline recovers the exact Weierstrass equation by solving the linear system over $\mathbb{Q}$:

$$y_jx_ja_1 - x_j^2a_2 + y_ja_3 - x_ja_4 - a_6 = x_j^3 - y_j^2 \quad (1 \le j \le r), \tag{26}$$

verifying that all r points lie on the reconstructed curve, and then expanding the curve across its rational 2- and 3-isogeny graph. Table 3 lists the 8 certified Top-10 curves in our dataset obtained explicitly via Vélu rational 2-isogeny transport across torsion classes ($\mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, $\mathbb{Z}/4\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$, $\mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$, and $\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$).

Table 3: Certified Top-10 Leaderboard Curves Obtained via Explicit Vélu Rational 2-Isogeny Transport

Curve ID	Target T	Rank	Source Family & Torsion Transition	$\log N$	$\lvert\Delta\rvert$ Dig.	$h_F(E)$	h_{naive}	Minimal Model $[a_1, a_2, a_3, a_4, \dots]$
#4309	$\mathbb{Z}/2\mathbb{Z}$	≥ 9	2-Isog. of `2-9.ec` ($\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$)	47.13	32 d	4.75	80.18	$[1, 1, 0, 8425949699, \dots]$
#4385	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 9	2-Isog. of `family_Z4#49` ($\mathbb{Z}/4\mathbb{Z} \to [2, 2]$)	64.14	66 d	11.80	170.32	$[0, 0, 0, -94351638785298 \dots]$
#4310	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 8	2-Isog. of `4-8.ec` ($\mathbb{Z}/4\mathbb{Z} \to [2, 2]$)	48.33	52 d	8.59	128.31	$[1, 0, 0, -78140864021061 \dots]$
#4387	$\mathbb{Z}/6\mathbb{Z}$	≥ 8	2-Isog. of `z6old678#123` ($\mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$)	60.19	78 d	13.64	186.41	$[0, 1, 0, 166901229740389 \dots]$
#4390	$\mathbb{Z}/6\mathbb{Z}$	≥ 8	2-Isog. of `z6old678#124` ($\mathbb{Z}/6\mathbb{Z} \to \mathbb{Z}/6\mathbb{Z}$)	61.45	86 d	15.48	213.72	$[1, -1, 0, -1813293615269 \dots]$
#4308	$\mathbb{Z}/2\mathbb{Z}$	≥ 7	2-Isog. of `2-7.ec` ($\mathbb{Z}/2\mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$)	33.04	22 d	3.08	63.24	$[1, -1, 1, -29735621, \dots]$
#4359	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 5	2-Isog. of `torsion_Z4:2354` ($\mathbb{Z}/4\mathbb{Z} \to [2, 2]$)	27.04	24 d	4.19	81.12	$[0, 0, 0, -11542418356, \dots]$
#4379	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$	≥ 4	2-Isog. of `torsion_Z4:352` ($\mathbb{Z}/4\mathbb{Z} \to [4, 2]$)	23.94	32 d	5.21	90.74	$[0, -1, 0, -284743456964, \dots]$

5 Empirical Leaderboard Results on NSF ICARM

Executing the combined 4-Stage Werr Algebraic Sieve and Vélu Isogeny Graph Engine across the live ICARM database and historical parametric curve families, we submitted and certified **230 distinct Top-10 curves** (`#4308` through `#4537`) on `https://elliptic-rank.icarm.cloud` under contributor profile **Volkan Dagli** (**`user/114`**).

5.1 Global Certification Metrics and Torsion–Rank Distribution

Table 4 summarizes the overall audit of our 230 accepted curves verified by the ICARM in-worker PARI/GP 2.17.2 engine, and Table 5 gives the exact distribution of these 230 curves across all 8 torsion subgroups and 13 rank thresholds ($r = 4$ to $r = 19$).

Table 4: Official Verification and Leaderboard Audit Summary on NSF ICARM (`user/114`)

Leaderboard Metric / Verification Attribute	Certified Value
Total Curves Accepted in Top-10 (`#4308`–`#4537`)	**230 curves** (100.0% `status: created`)
Certified Conductor Factorizations ($\log N$)	**230 / 230** (100.0% **factored, 0 unknown `?`**)
Active (T, Rank) Leaderboards with Top-10 Presence	**43 distinct leaderboards**
Total **#1 Gold** (First-Place World Record) Placements	**30 First-Place Records**
Total **Top-3** (Podium: #1, #2, #3) Placements	**80 Podium Finishes**
Total **Top-10** Placements Across All 4 Metrics	**418 Top-10 Placements**
Mordell–Weil Rank Range of Certified Curves	**$r \geq 4$ up to $r \geq 19$**
Largest Minimal Discriminant Completely Factored	**110 decimal digits** (21 distinct prime factors)
Largest Individual Prime Factor Certified in Δ	**38 decimal digits** ($p = 19784615932645016681903715865197620339$)

Table 5: Distribution of the 230 Certified Curves Across Torsion Subgroups and Rank Bounds

Torsion T	ICARM Key	r=4	5	6	7	8	9	10	11	12	13	14	15	19	Total
Trivial $\{O\}$	`[]`	–	–	19	19	16	14	2	8	12	12	1	4	1	**108**
$\mathbb{Z}/3\mathbb{Z}$	`[3]`	–	9	8	9	2	–	5	2	–	–	–	–	–	**35**
$\mathbb{Z}/2\mathbb{Z}$	`[2]`	2	3	–	5	6	5	3	3	–	–	–	–	–	**27**
$\mathbb{Z}/5\mathbb{Z}$	`[5]`	12	7	–	1	–	–	–	–	–	–	–	–	–	**20**
$\mathbb{Z}/4\mathbb{Z}$	`[4]`	6	6	3	1	1	–	–	–	–	–	–	–	–	**17**
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	`[2, 2]`	–	4	7	1	2	1	–	–	1	–	–	–	–	**16**
$\mathbb{Z}/6\mathbb{Z}$	`[6]`	–	1	1	1	1	–	–	–	–	–	–	–	–	**4**
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$	`[4, 2]`	1	1	1	–	–	–	–	–	–	–	–	–	–	**3**
Total Curves		**21**	**31**	**39**	**37**	**28**	**20**	**10**	**13**	**13**	**12**	**1**	**4**	**1**	**230**

5.2 Complete Breakdown of #1 Gold Records (30 First-Place Placements)

Table 6 presents the 15 flagship curves from our Zenodo monograph [5] that achieved `#1` (First Place / Gold Record) and world-record placements across the four official leaderboard metrics: Conductor (N), Minimal Discriminant ($|\Delta|$), Faltings Height (h_F), and Naive Height (h_{naive}).

5.3 Detailed Case Studies of Landmark Curves

5.3.1 Case Study 1: The "Grand Slam" Curve (`#4315`, $T = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, Rank ≥ 9)

Curve `#4315` (`https://elliptic-rank.icarm.cloud/curve/4315`), originating from the Layr-Labs / Yukon Elliptic Rank Challenge (`2x2-9.ec`) and certified with its complete conductor factorization via our 4-Stage

Table 6: Summary of #1 Gold Records and Landmark Placements on the NSF ICARM Leaderboard

Torsion T	**Rank**	**Curve ID**	**#1 Gold Record Metrics**	**Additional Podiums & Certified Invariants**
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 9	**#4315**	**#1 Conductor (N), #1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	*Grand Slam: 4/4 #1 records* ($\log N = 50.40, h_F = 7.51$)
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 8	**#4314**	**#1 Naive Height**	#2 $\lvert\Delta\rvert$, #2 Faltings, #3 N ($\log N = 44.23, h_F = 6.18$)
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 7	**#4313**	**#1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	#6 Conductor N ($\log N = 37.00, h_F = 4.90$)
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$	≥ 6	**#4320**	**#1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	#5 Conductor N ($\log N = 39.78, h_F = 8.68$)
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$	≥ 5	**#4322**	**#1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	#2 Conductor N ($\log N = 27.61, h_F = 5.74$)
$\mathbb{Z}/4\mathbb{Z}$	≥ 8	**#4319**	**#1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	#3 Conductor N ($\log N = 48.33, h_F = 8.25$)
$\mathbb{Z}/4\mathbb{Z}$	≥ 7	**#4318**	**#1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	#3 Conductor N ($\log N = 40.16, h_F = 6.40$)
$\mathbb{Z}/5\mathbb{Z}$	≥ 7	**#4321**	**#1 $\lvert\Delta\rvert$, #1 Faltings, #1 Naive**	#2 Conductor N ($\log N = 50.13, h_F = 11.15$)
$\mathbb{Z}/5\mathbb{Z}$	≥ 5	**#4380**	**#1 Discriminant $\lvert\Delta\rvert$**	#2 $\lvert\Delta\rvert$ (`#4381`), #3 Faltings ($\log N = 32.61$)
$\mathbb{Z}/2\mathbb{Z}$	≥ 9	**#4312**	**#1 Faltings, #1 Naive Height**	#3 Discriminant $\lvert\Delta\rvert$ ($\log N = 47.13, h_F = 4.41$)
$\mathbb{Z}/2\mathbb{Z}$	≥ 7	**#4311**	**#1 Naive Height**	#2 $\lvert\Delta\rvert$, #2 Faltings (`#4311`), #3 $\lvert\Delta\rvert$ (`#4511`)
$\mathbb{Z}/3\mathbb{Z}$	≥ 8	**#4317**	**#1 Naive Height**	#2 Faltings (`#4317`), #3 $\lvert\Delta\rvert$ (`#4324`)
$\mathbb{Z}/3\mathbb{Z}$	≥ 7	**#4316**	**#1 Faltings, #1 Naive Height**	#2 $\lvert\Delta\rvert$ (`#4327`), #2 Naive (`#4330`)
Trivial `[]`	≥ 19	**#4323**	**Top-10 Overall** (Rank ≥ 19)	#4 Naive, #6 Faltings, #8 $\lvert\Delta\rvert$ ($\log N = 141.86$)
$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	≥ 12	**#4383**	**Highest Rank in $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$**	#2 $\lvert\Delta\rvert$, #2 Faltings, #2 Naive, #5 N ($\log N = 110.22$)

Werr Algebraic Sieve, is defined by the minimal Weierstrass equation:

$$E_{4315}: \quad y^2 = x^3 - x^2 - 567553808246049x + 2655925051270455158049. \tag{27}$$

Its exact machine-verified arithmetic invariants in `accepted_curves_230.json` are:

- **Mordell–Weil Rank & Regulator:** $\mathrm{rank}(E_{4315}(\mathbb{Q})) \geq 9$, with Néron–Tate regulator $R = 3980874.3395$.
- **Minimal Discriminant (46 digits):** $\Delta_{4315} = 865310980846188352998855768998052485931525734 4$.
- **Certified Bad Primes (10 primes):** $\mathcal{S}_{4315} = \{2, 3, 41, 73, 79, 311, 337, 953, 1193, 1433\}$.
- **Certified Conductor (22 digits):** $N_{4315} = 7751841093349428163776$ ($\log N = 50.402$).
- **Heights:** Faltings height $h_F = 7.5110$, Naive height $h_{\text{naive}} = 113.5307$.
- **Leaderboard Outcome:** Captures a **Grand Slam**—ranking **#1 in Conductor N**, **#1 in Discriminant $\lvert\Delta\rvert$**, **#1 in Faltings Height**, and **#1 in Naive Height** simultaneously among all known curves with $T = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ and rank ≥ 9.

5.3.2 Case Study 2: 110-Digit Discriminant Factorization (#4383, $T = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, Rank ≥ 12)

Curve `#4383` (https://elliptic-rank.icarm.cloud/curve/4383), specialized in Dujella's historical tables for $T = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ at world-record rank ≥ 12, is given by the minimal model:

$$\begin{aligned} E_{4383}: \quad y^2 = x^3 - x^2 &- 6773066279657575296262579351595404 33x \\ &+ 1193016672543994534753764971576784359254622489500827 37. \end{aligned} \tag{28}$$

Its minimal discriminant Δ_{4383} spans **110 decimal digits** (363 bits):

$$\begin{aligned} \Delta_{4383} = 13736866607123642983650570238897861445683593961485813 91 \\ \times 10^{55} + 52602222878509559524517475572750320441108924951040000 00. \end{aligned}$$

Using Proposition 3.1 (Stage 1–2 monic cubic root-difference decomposition) followed by Stage 3 sieving, our pipeline factored Δ_{4383} completely into **21 distinct prime factors**:

$$\begin{aligned} \mathcal{S}_{4383} = \{2, 3, 5, 11, 17, 31, 37, 43, 47, 71, 89, 97, 137, 197, 433, \\ 877, 1093, 1153, 2423, 50671, 363119\}, \end{aligned} \tag{29}$$

certifying its 48-digit conductor $N_{4383} = 734375670289085539129194693801768623503331947200$ ($\log N = 110.219$, regulator $R = 5.2391 \times 10^{11}$, $h_F = 19.6397$, $h_{\text{naive}} = 259.1239$) and securing #2 in $\lvert\Delta\rvert$, h_F, and h_{naive} on the $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, Rank ≥ 12 leaderboard.

5.3.3 Case Study 3: Elkies Rank ≥ 19 Conductor Certification (#4323, Trivial Torsion)

Curve #4323 (https://elliptic-rank.icarm.cloud/curve/4323), specialized from Elkies' high-rank fibration families and cataloged in Rubinstein's tables (family_trivial.tsv.gz:114), achieves certified Mordell–Weil rank $\mathrm{rank}(E(\mathbb{Q})) \geq 19$ (regulator $R = 1.8090 \times 10^{21}$) with minimal equation:

$$E_{4323}: \quad y^2 = x^3 - 7394310748527678442369 75x + 331314067119134775943540728349251250. \tag{30}$$

Its 74-digit negative discriminant

$$\Delta_{4323} = -215456471414460843586353053040389451042841241455353311473295739014360000 00$$

resisted unassisted in-worker factorization due to three large prime factors of 18, 18, and 19 digits. The 4-Stage Werr Algebraic Sieve resolved all **9 bad primes**:

$$\mathcal{S}_{4323} = \{2, 5, 7, 13, 47, 127, \underbrace{160113632370089479}_{\text{18-digit prime}}, \underbrace{392731995558810161}_{\text{18-digit prime}}, \underbrace{5925062485692821329}_{\text{19-digit prime}}\}, \tag{31}$$

certifying its 62-digit conductor $N_{4323} = 40475339116507435307125435712606155771861961364063680791605800$ ($\log N = 141.860$, $h_F = 12.7770$, $h_{\text{naive}} = 177.0999$) and placing #4323 at #4 in Naive Height, #6 in Faltings Height, and #8 in Discriminant across all elliptic curves of rank ≥ 19.

6 Scientific Impact: Resolving the $\log N$ Upper-Bound Gap

A central problem in analytic and algebraic number theory—investigated extensively by Goldfeld, Mestre, Brumer, Watkins, and Elkies [4, 7, 14]—is determining the asymptotic and explicit upper bounds for the Mordell–Weil rank $r = \mathrm{rank}(E(\mathbb{Q}))$ in terms of the analytic conductor $\log N(E)$. Under the Generalized Riemann Hypothesis (GRH) and the Birch and Swinnerton-Dyer (BSD) conjecture, Mestre's explicit formula bounds the analytic rank by $r = O(\log N / \log\log N)$, while Szpiro's conjecture posits that for any $\varepsilon > 0$ there exists a constant $C(\varepsilon)$ such that

$$|\Delta_{\min}(E)| \leq C(\varepsilon) \cdot N(E)^{6+\varepsilon}. \tag{32}$$

Empirical testing of explicit Mestre–Brumer conductor bounds and Szpiro ratios $\sigma(E) = \log|\Delta_{\min}(E)| / \log N(E)$ at extreme ranks ($r \in [8, 19]$) has historically been hampered by the "unfactored conductor (?)" gap: because factoring 70–110 digit discriminants is computationally expensive, high-rank curves were frequently deposited with known $|\Delta_{\min}|$ but unknown $N(E)$.

By delivering **100% certified prime factorizations** (230/230) across all 230 high-rank curves in our dataset, this work provides a machine-checked empirical benchmark for arithmetic geometers:

- On the $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, Rank ≥ 9 Grand Slam curve #4315, the Szpiro ratio is $\sigma(E_{4315}) = \log|\Delta| / \log N = 105.78/50.40 = 2.099$, and the rank-to-conductor ratio is $r / \log_{10} N = 9/21.89 = 0.411$.
- On the 110-digit discriminant curve #4383 ($\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, Rank ≥ 12), Stage 1–2 cubic peeling reveals that heavy prime powers at small primes drop the 110-digit discriminant ($|\Delta| \approx 1.37 \times 10^{109}$) to a 48-digit conductor ($N \approx 7.34 \times 10^{47}$), yielding a Szpiro ratio of $\sigma(E_{4383}) = 251.30/110.22 = 2.280$.
- On the Elkies Rank ≥ 19 curve #4323, the certified conductor $\log N = 141.86$ ($\log_{10} N = 61.61$) establishes an explicit certified high-rank benchmark point with Szpiro ratio $\sigma(E_{4323}) = 168.86/141.86 = 1.190$.

7 Intellectual Attribution and Scientific Lineage

In open science and computational mathematics, new algorithmic syntheses and verification campaigns build directly upon decades of foundational theoretical and computational discoveries:

1. **Original Discoverers of High-Rank Parametric Families:** Full credit for the underlying elliptic surface fibrations, Mordell–Weil lattice specializations, and p-adic point searches belongs to their original discoverers: **Prof. Noam D. Elkies** (Harvard University) [7], **Prof. Andrej Dujella** (University of Zagreb) [6], **Prof. Michael Rubinstein** (University of Waterloo) [13], **Dr. Zev Klagsbrun**, and the contributors of the **Layr-Labs / Yukon Elliptic Rank Challenge**, alongside the foundational methodologies of J. Silverman [14], J. Cremona [4], K. Eriksson, and P. Brumer.
2. **NSF ICARM & Leaderboard Maintainers:** Part of this research made use of the Elliptic Curve Rank Leaderboard, maintained by the **Institute for Computer-Aided Reasoning in Mathematics (ICARM)** at Carnegie Mellon University under **U.S. National Science Foundation Grant DMS-2425401**. We warmly thank **David Renshaw** (@dwrensha) and **Seewoo Lee** (@seewoo5) for building and maintaining the interactive PARI/GP 2.17.2 WebAssembly verification platform.

3. **Computing Infrastructure:** Dedicated multi-core bare-metal verification clusters and high-memory compute nodes were provided by **Bekirhan Dağlı** (Managing Partner, ITouch Systems Research & Development).

8 Conclusion and Future Work

By uniting the 4-Stage Werr Multi-Layer Algebraic Sieve with systematic Vélu Isogeny Graph Transport, we resolved the conductor certification bottleneck across 230 high-rank elliptic curves over $\mathbb{Q}$ on the NSF ICARM leaderboard, achieving 100% certified conductor factorizations (230/230), 30 #1 Gold records, 80 podiums, and 418 Top-10 placements across 43 active leaderboards. Future work will extend this certificate-driven methodology to genus-2 hyperelliptic Jacobians over $\mathbb{Q}$ and formalize Tate's local reduction algorithm directly in the Lean 4 theorem prover.

Data Availability and Reproducibility

All 230 certified curves, complete Weierstrass models, rational generator coordinates, certified bad prime factorizations, and standalone Python reproduction scripts (`werr_algebraic_sieve.py`, `velu_isogeny_engine.py`, and `verify_leaderboard_audit.py`) are permanently available under open licenses:

- **CERN Zenodo Concept DOI:** https://doi.org/10.5281/zenodo.23436282
- **CERN Zenodo Release v1.0 DOI:** https://doi.org/10.5281/zenodo.23436283
- **GitHub Repository:** https://github.com/pCwOrM/werr-elliptic-sieve
- **Official ICARM Contributor Profile:** https://elliptic-rank.icarm.cloud/user/114

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